

Consider play tennis dataset:

Total samples = 14.

Yes = 9

No = 5.

Calculating Entropy & IG.

Used in ID3.

Entropy Formula:

$$\text{Entropy}(S) = -P_1 \log(P_1) - P_2 \log(P_2)$$

Step 1: Entropy For entire dataset:

where: $p(\text{yes}) = 9/14$, $p(\text{no}) = 5/14$

$$\therefore \text{Entropy}(S) = (9/14) \log(9/14) - (5/14) \log(5/14)$$

$$\log(9/14) \approx -0.64, \quad \log(5/14) \approx -1.048.$$

$$\therefore \text{Entropy}(S) = -(0.64 \times 9/14) - (1.048 \times 5/14)$$

$$\approx 0.94.$$

$$\boxed{\therefore \text{Initial entropy} = 0.94}$$

Step 2: Calculate IG for "outlook".

Outlook values:

outlook	yes	no	Total
sunny	2	3	5
overcast	4	0	4
Rain	3	2	5

2.1. Entropy for sunny:

$$\text{yes} = 3, \text{ no} = 2.$$

$$\begin{aligned}\text{Entropy} &= (-2/5) \log(2/5) - (3/5) \cdot \log(3/5) \\ &\approx 0.97.\end{aligned}$$

2.2. Entropy for overcast

$$\text{yes} = 4, \text{ no} = 0.$$

$$\text{pure node: Entropy} = 0.$$

2.3. Entropy for Rain

$$\text{yes} = 3, \text{ no} = 2.$$

$$\therefore \text{Entropy} \approx 0.97.$$

Step 3: Weighted Entropy.

$$\begin{aligned}\text{Weighted Entropy} &= \text{Sunny} + \text{Overcast} + \text{Rain} \\ &= (5/14 \times 0.97) + (4/14 \times 0) + (5/14 \times 0.97) \\ &\approx 0.69.\end{aligned}$$

Step 4: Information Gain:

$$\begin{aligned}\text{IG} &= \text{Entropy}(\text{parent}) - \text{Weighted Entropy} \\ &= 0.94 - 0.69 = 0.25.\end{aligned}$$

Step 5: Repeat the steps for other features.

<u>Features</u>	<u>IG</u>
1. Outlook	0.25
2. Humidity	0.151
3. Wind	0.048.
4. Temperature	0.029.

— which ever the IG is high, will be the

Root node.

∴ Outlook is first Root node.

— For selecting the next root node repeat the same process for remaining features.

USING GINI INDEX.

Used in CART (Classification and Regression Trees)

Step 1: Gini of Dataset:

$$\begin{aligned}\text{Formula: } Gini &= 1 - (P_1^2 + P_2^2) \\ &= 1 - ((9/14)^2 + (5/14)^2) \\ &= 1 - (81/196 + 25/196) \\ &= 1 - 106/196 \\ &= 0.459.\end{aligned}$$

Step 2: Gini for Outlook split:

Sunny: yes=2, no=3.

$$\begin{aligned}Gini &= 1 - ((2/5)^2 + (3/5)^2) \\ &= 1 - (0.16 + 0.36) \\ &= 0.48\end{aligned}$$

Overcast: yes=4, no=0

$$\therefore \text{Gini}^{\circ} = 0.$$

Rain : yes = 3, no = 2.

$$\text{Gini}^{\circ} = 0.48.$$

Step 3: Weighted Gini.

$$\begin{aligned} \text{Weighted Gini}^{\circ} &= \text{Sunny} + \text{Overcast} + \text{Rainy} \\ &= (5/14 \times 0.48) + (4/14 \times 0) + (5/14 \times 0.48) \\ &= 0.34. \end{aligned}$$

Step 4: Gini Gain:

$$\begin{aligned} \text{Gain} &= \text{Gini}(\text{parent}) - \text{Weighted Gini} \\ &= 0.459 - 0.34 = 0.119. \end{aligned}$$

Feature composition Using Gini:

<u>Features</u>	<u>Gini</u>
Outlook	0.34 (lowest)
Humidity	0.37
Wind	0.43.

Temperature

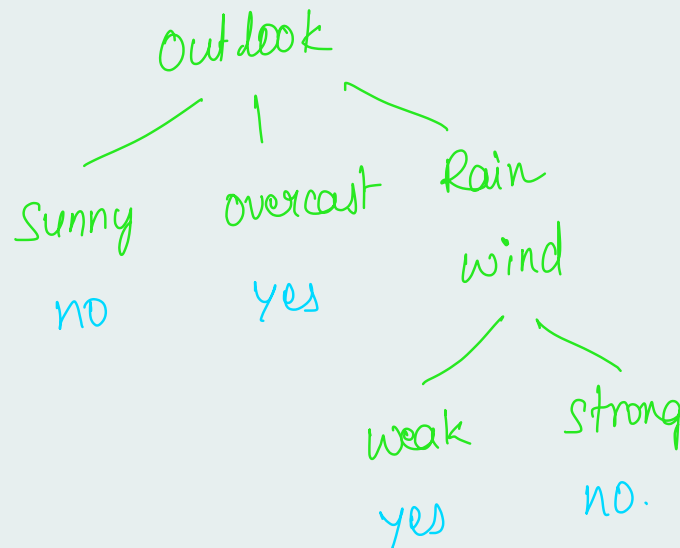
0.44

Gini $\rightarrow$ Impurity in dataset

Lowest Gini $\rightarrow$ Best split

Root node = outlook

Final Tree :



High entropy $\rightarrow$ Mixed data

Low entropy $\rightarrow$ Pure data

Decision tree $\rightarrow$ Find split that reduce entropy.

Decision Tree for continuous data :

Age purchase

25 yes

33 yes

17 no

55 no

Step 1 : Sort the data.

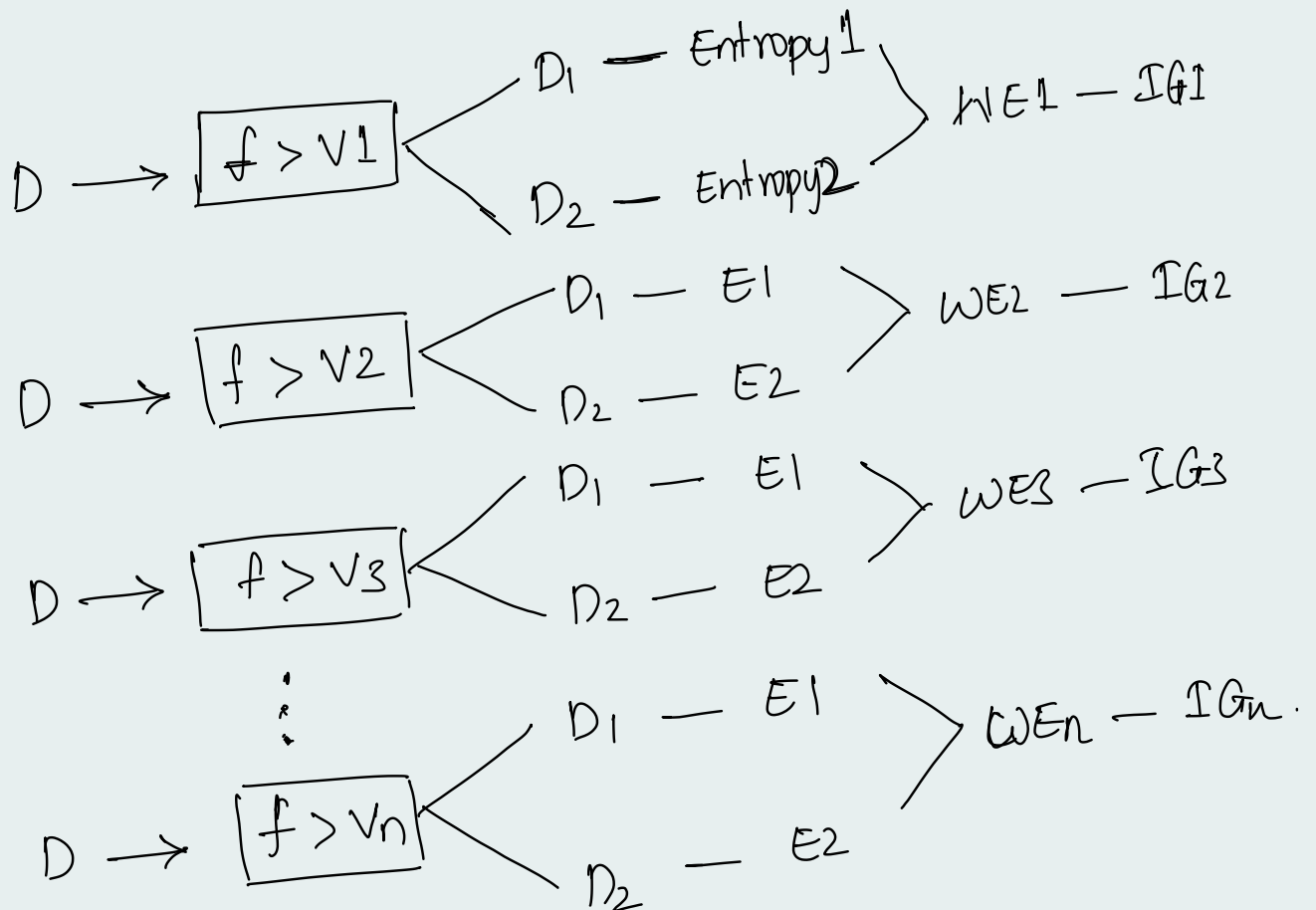
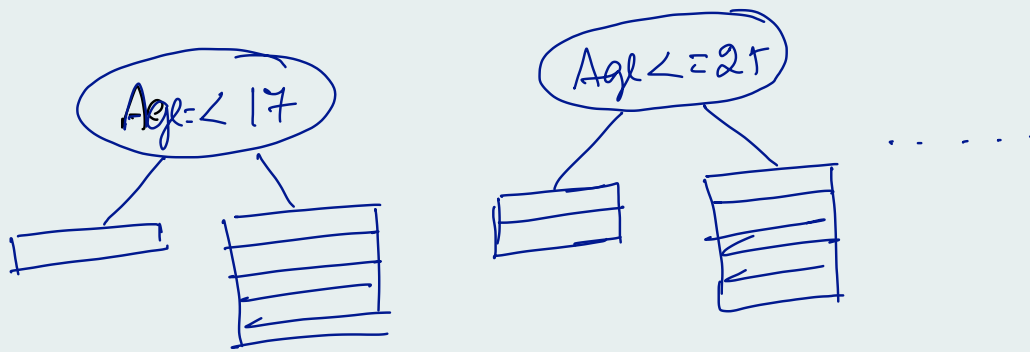
Step 2 : Split the data
on each data pt
vs rest of data.

73

no

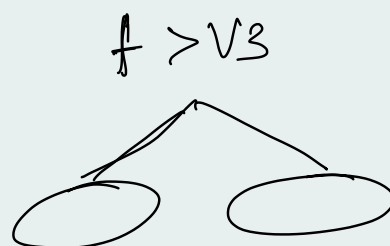
26

yes

 $\therefore$ Total no. of splits = 'n'

— Find the maximum IG.

Let's say IG3 highest value. SO split will start



Several algorithms exist to build DT

1. ID3 (Iterative Dichotomiser 3)

- It will select features with highest IG.
- Metrics used
 - Entropy and IG.

Characteristics

1. Uses entropy
2. works mainly with categorical features.
3. Creates multi branch splits
4. No pruning mechanism

Limitations:

1. Can not handle continuous variables well
2. Can overfit
3. Sensitive to missing values.

2. C4.5

- C4.5 is an improved version of ID3
- metric used Gini Ratio.

$$\text{Gini Ratio} = \frac{IG}{\text{Split Information}}$$

Why Gini Ratio:

IG tends to favour features with many unique values. For ex.

Feature 'ID' would have high IG but useless.

— Gini ratio penalizes such split.

Features:

1. Handle continuous data
2. Handle missing data
3. Uses tree pruning

3. CART

— most widely used DT algo in practice

— Libraries like sklearn also uses CART.

Metric used:

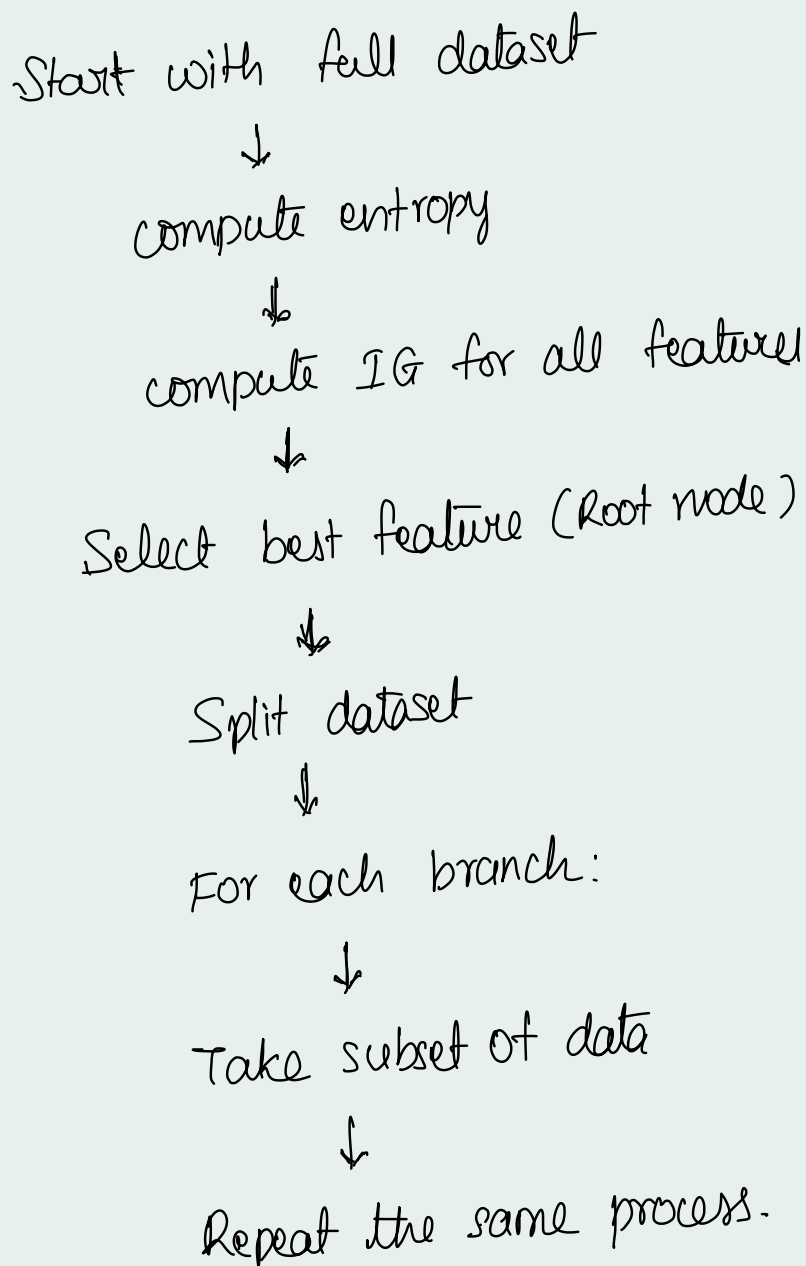
$$\text{Gini} = 1 - \sum p_i^2$$

Key characteristics:

1. Produces binary split only

2. Works for classification & Reg
3. Uses Gini index.
4. Supports pruning.

Visual Flow of algorithm:



Handling missing values in DT:

Some algorithms like C4.5 & CART can manage

missing values internally.

method 1: Ignore missing values during split calculation

- Algo simply ignores rows where the feature values are missing.

method 2: Probabilistic split (Used in C4.5)

- Instead of ignoring the record, the algorithm distributes the sample probabilistically.

method 3: Surrogate splits (Used in CART)

- If primary split feature is missing, CART looks for another feature that behaves similarly.

For Ex: Age ≤ 30 , But Age is missing.

Tree may use income ≤ 50000

Bcz income column correlates with age.

- This is called surrogate split.

